9/28/25, 2:35 PM Rich Text Editor, textQuestionEditor

25AIMLPCC302IAE-II 04 OCT 2026 Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Artificial Intelligence and Machine Learning - SEMESTER III

25AIMLPCC302 - Data Structure

Time : 1 Hour **Maximum Marks : 20**

St. John College of Engineering & Management SJCEM PALGHAR

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	BLT Level
	a)A full binary tree is a tree in which every node has: _____ child/children	1	CO4	L2
	b)Topological sorting provides an ordering of vertices such that: A. Every vertex has the same degree B. For every directed edge U → V, U appears before V C. V always appears before U D. Vertices are arranged alphabetically	1	CO5	L2
	c)A good hash function should ideally distribute keys: A. At the same location B. Uniformly across the hash table C. Only at even locations D. Only at the first location	1	CO6	L2
	d)Construct a Binary Search Tree by inserting the following elements in the given order: 50, 30, 70, 20, 40, 60, 80	1	CO4	L3
	e)DFS primarily uses which data structure for maintaining the traversal order? A. Queue B. Stack C. Heap D. Priority Queue	1	CO5	L2
	f)A list contains the following elements: 15, 8, 22, 10, 30 To locate the element 22 without rearranging the list, which technique is most appropriate? A. Linear Search B. Binary Search C. Hashing D. Tree traversal	1	CO6	L3
Q2	a)Explain how B-trees are useful in database indexing.	5	CO4	L1
	OR			
	b)Generate Huffman code for the letters a, b, c, d, e, f having frequencies 16, 8 4, 2, 1, 1.	5	CO4	L3

Q3	a)Co.ıstruct adjacency matrix and adjacency list for the graph:	5	CO5	L4
	OR			
	b)Perform BFS on the graph.	5	CO5	L3
Q4	a)Discuss various collision resolution techniques in hashing with suitable examples.	5	CO6	L3
	OR			
	b)Sort the array [20, 12, 10, 15, 2, 38, 72, 26, 55, 92, 63] and search 72 using binary search	5	CO6	L3